NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 0

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total marks: 10 X 1= 10**

QUESTION 1:

Which of the following allows us to identify objects and extract information?

a. RFID
b. Sensors
c. Actuators
d. IoT Nodes

Correct Answer: a. RFID

Detailed Solution: RFID Technology allows us to automatically identify and track tags that are attached to the objects. It extracts information from the tags through electromagnetic fields.
See lecture 1 @ 12:57

QUESTION 2:

How many layers does Zigbee consist of?

a. 1
b. 2
c. 3
d. 4

Correct Answer: d. 4
Detailed Solution: Zigbee consists of 4 layers: Physical, Medium Access Control, Network, and Application.
See lecture 48 @ 16:11

QUESTION 3:

Which of the following is not a component of cloud computing?

a. Clients
b. Local Servers
c. Services
d. Applications

Correct Answer: b. Local Servers

Detailed Solution: Cloud computing components include clients, services, applications, platform, storage, and infrastructure.
See lecture 37 @ 23:29

QUESTION 4:

Which of the following is a distance measuring sensor module?

a. DHT22
b. HC-SR04
c. TSL2591
d. HC-SR505

Correct Answer: b. HC-SR04

Detailed Solution: HC-SR04 is the distance measuring module ultrasonic sensor, which measures the distance between 2cm~450cm.
See lecture 3 @ 5:00

QUESTION 5:

Which of the following is a component in a typical sensor network?

a. Sink
b. Gateway
c. Router
d. All of these

Correct Answer: d. All of these

Detailed Solution: A typical sensor network comprises of sensor nodes, routers, gateway, and sink.

QUESTION 6:

Which of the following sensors are responsible for measuring orientation and angular velocity?

a. Accelerometer
b. GPS
c. Temperature
d. None of these

Correct Answer: d. None of these

Detailed Solution: A gyroscope is responsible for measuring orientation and angular velocity.

See lecture 59 @ 15:41

QUESTION 7:

"ISA 100.11A" is a wireless networking technology standard. ISA stands for _____.

a. International Society of Automation
b. International Society of Advancement
c. Industrial Society of Automation
d. Industrial Society of Advancement

Correct Answer: a. International Society of Automation

Detailed Solution: ISA100.11a is a wireless networking technology standard developed by the International Society of Automation (ISA).
See lecture 13@ 15:55

QUESTION 8:

Which of the following is not a difference between traditional data center and cloud computing?

a. Scalability
b. Flexibility
c. Elasticity
d. Storage

Correct Answer: d. Storage

Detailed Solution: Major differences between traditional data center and cloud computing include scalability, flexibility, elasticity, automation, running costs, and security
See lecture 39 @ 11:02

QUESTION 9:

Smart grid is also known as the energy internet.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Smart grid is also known as the energy internet.
See lecture 51 @ 7:51

QUESTION 10:

Can a point of node failure result in the partition of the network in the stationary sensor network?

a. Yes

b. No

Correct Answer: a. Yes

Detailed Solution: If there is a failure in the stationary sensor network then it is likely that the point of failure can partition the network into two or more fragments.

See lecture 18 @ 01:10

**********END**********

Introduction to Internet of Things

Assignment-Week 1

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

What is the full form of IoT?

a. Internet of Tasks
b. Internet of Things
c. Internet of Tasks
d. None of these

Correct Answer: b. Internet of Things

Detailed Solution: The full form of IoT is "Internet of Things"

See lecture 1 (Introduction to IoT – Part - I) @ 1:30

QUESTION 2:

Which of the following technologies have unified and has resulted in the evolution of IoT?

a. Low-power embedded systems
b. Cloud Computing
c. Machine Learning
d. All of these

Correct Answer: d. All of these

Detailed Solution: Unification of technologies which has resulted in the advancement of IoT are –

a. Low-power embedded systems
b. Cloud Computing
c. Big Data
d. Machine Learning
e. Networking

See lecture 1 (Introduction to IoT – Part - I) @ 5:54

QUESTION 3:

Which of the following are the enablers of IoT?

a. RFID
b. Nanotechnology
c. Sensors
d. All of these

Correct Answer: d. All of these

Detailed Solution: The enables of IoT are –

a. RFID
b. Nanotechnology
c. Sensors
d. Smart Networks

See lecture 2 (Introduction to IoT – Part - I) @ 12:50

QUESTION 4:

Which of the following is/are a function of an IoT Gateway?

a. Connects the IoT LAN to a WAN
b. Can implement several LAN and WAN
c. Forwards packets between LAN and WAN on the IP layer
d. All of these

Correct Answer: d. All of these

Detailed Solution: The functionalities of an IoT Gateway are –

a. Connects the IoT LAN to a WAN
b. Can implement several LAN and WAN
c. Forwards packets between LAN and WAN on the IP layer

See lecture 2 (Introduction to IoT – Part - II) @ 3:09

QUESTION 5:

State whether the following statement is True or False.

Statement: "The same address of an IoT device may be repeated in the domain of another gateway. The gateway has a unique network prefix, which can be used to identify them globally."

a. True
b. False

Correct Answer: a. True

Detailed Solution: The same address of an IoT device may be repeated in the domain of another gateway. The gateway has a unique network prefix, which can be used to identify them globally.

See lecture 2 (Introduction to IoT – Part - II) @ 07:00

QUESTION 6:

In _____ a node/network is connected to multiple networks for improved reliability.

a. Transparent roaming
b. Multi-homing
c. None of these
d. Both (a) and (b)

Correct Answer: b. Multi-homing

Detailed Solution: In multi-homing, a node/network is connected to multiple networks for improved reliability.

See lecture 2 (Introduction to IoT – Part - II) @ 15:22

QUESTION 7:

The IPv4 notation uses _____ number of bits to represent an address.

a. 33
b. 32
c. Both (a) and (b)
d. None of these

Correct Answer: b. 32

Detailed Solution: The IPv4 notation uses 32 bits to represent an address.

See lecture 2 (Introduction to IoT – Part - II) @ 16:33

QUESTION 8:

The "Source Address" field in the IPv6 header is of _____ bits in length.

a. 32

b. 64

c. 128

d. None of these

Correct Answer: c. 128

Detailed Solution: The IPv6 header uses 128 bits to represent the "Source Address".

See lecture 2 (Introduction to IoT – Part - II) @ 18:03

QUESTION 9:

A device which detects or measures a physical property and records, indicates, or otherwise responds to it is called _____?

a. A sensor

b. An actuator

c. A Transducer

d. A processor

Correct Answer: a. A sensor

Detailed Solution: A device which detects or measures a physical property and records, indicates, or otherwise responds to it is called a sensor.

See lecture 3 (Sensing) @ 03:28

QUESTION 10:

A microphone is an example of an _____.

a. Input device and actuator
b. Only actuator
c. Only Transducer
d. Input device and transducer

Correct Answer: d. Input device and transducer

Detailed Solution: A microphone is an example of an input device and transducer.

See lecture 3 (Sensing) @ 10:31

QUESTION 11:

State whether the following statement is True or False.

Statement: A sensor is only sensitive to the measured property for which it has been made (e.g., A temperature sensor senses the ambient temperature of a room.)

a. True
b. False

Correct Answer: a. True

Detailed Solution: A sensor is only sensitive to the measured property (e.g., A temperature sensor senses the ambient temperature of a room.)

.
See lecture 3 (Sensing) @ 12:11

QUESTION 12:

Based on the data type, sensors are classified as _____.

a. Only scalar
b. Only vector
c. Scalar and Vector/Multimedia
d. None of these

Correct Answer: c. Scalar and Vector/Multimedia

Detailed Solution: Based on the data type, sensors are classified as Scalar and Vector/Multimedia.

See lecture 3 (Sensing) @ 13:05

QUESTION 13:

State whether the following statement is True or False.

Statement: The more the resolution of a sensor, the less accurate its precision.

a. True
b. False

Correct Answer: b. False
Detailed Solution: The more the resolution of a sensor, the more accurate its precision.

See lecture 3 (Sensing) @ 12:46

QUESTION 14:

The sensitivity of a sensor under real conditions may differ from the value specified. This is called _____.

a. Physical error
b. Sensitivity error
c. Rounding error
d. None of these

Correct Answer: b. Sensitivity error

Detailed Solution: The sensitivity of a sensor under real conditions may differ from the value specified.

See lecture 3 (Sensitivity error) @ 20:56

QUESTION 15:

A sensor node is made up of which of the following?

a. Sensor/Sensing units
b. A processing unit
c. A power unit
d. All of these

Correct Answer: d. All of these

Detailed Solution: A sensor node is made up of a combination of sensor/sensors, a processor unit, a radio unit, and a power unit.

See Page number – 101, Chapter - 5, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

**********END**********

Introduction to Internet of Things

Assignment-Week 2

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

The full form of MQTT is

a. Message Query Telemetry Transport

b. Message Queue Telemetry Transport

c. Message Queue Telemedicine Transport

d. None of these

Correct Answer: b. Message Queue Telemetry Transport

Detailed Solution: The full form of MQTT is Message Queue Telemetry Transport.

See lecture 6 (Basics of IoT Networking – Part II) @ 01:51

QUESTION 2:

In MQTT, a _____ controls the publish-subscribe messaging pattern.

a. Message Broker

b. Publishers

c. Subscribers

d. All of these

Correct Answer: a. Message Broker

Detailed Solution: In MQTT, a message broker controls the publish-subscribe messaging pattern.

See lecture 6 (Basics of IoT Networking – Part II) @ 03:33

QUESTION 3:

Which of the following is NOT a Data Protocol in IoT?

a. MQTT
b. CoAP
c. Websocket
d. None of these

Correct Answer: d. None of these

Detailed Solution: The data protocols in IoT are –

a) MQTT

b) CoAP

c) AMQP

d) Websocket

See lecture 6 (Basics of IoT Networking – Part II) @ 01:16

QUESTION 4:

In MQTT for IoT, the Publishers are _____.

a. Lightweight sensors
b. Actuators
c. Processing units
d. None of these

Correct Answer: a. Lightweight sensors

Detailed Solution: In MQTT, the Publishers are lightweight sensors.

See lecture 6 (Basics of IoT Networking – Part II) @ 04:49

QUESTION 5:

Which of the following is NOT a method in MQTT?

a. Break
b. Connect
c. Disconnect
d. None of these

Correct Answer: a. Break

Detailed Solution: Following are the methods in MQTT –

a. Connect
b. Disconnect
c. Subscribe
d. Unsubscribe
e. Publish

See lecture 6 (Basics of IoT Networking – Part II) @ 06:18

QUESTION 6:

Which of the following protocols in IoT have been designed for Machine to Machine (M2M) applications such as smart energy and building automation?

a. MQTT
b. CoAP
c. Websocket
d. All of these

Correct Answer: b. CoAP

Detailed Solution: CoAP has been designed for Machine to Machine (M2M) applications such as smart energy and building automation.

See lecture 7 (Basics of IoT Networking – Part III) @ 00:41

QUESTION 7:

Similar to HTTP, CoAP utilizes which of the following for operation?

a. GET
b. PUT
c. PUSH
d. All of these

Correct Answer: d. All of these

Detailed Solution: Similar to HTTP, CoAP utilizes GET, PUT, PUSH, DELETE message requests to retrieve, create, update, and delete, respectively.

See lecture 7 (Basics of IoT Networking – Part III) @ 08:24

QUESTION 8:

What is the full form of XMPP in IoT?

a. Extensible Markup Page Processing
b. Extensible Messaging and Presence Protocol
c. Both (a) and (b)
d. Neither (a) nor (b)

Correct Answer: b. Extensible Messaging and Presence Protocol

Detailed Solution: XMPP stands for Extensible Messaging and Presence Protocol.

See lecture 7 (Basics of IoT Networking – Part III) @ 11:58

QUESTION 9:

State whether the following statement if True or False.

Statement: XMPP is an open standard protocol.

a. True
b. False

Correct Answer: a. True

Detailed Solution: XMPP is an open standard protocol.

See lecture 7 (Basics of IoT Networking – Part III) @ 11:58

QUESTION 10:

What is the full form AMQP in context of IoT protocol?

a. Another Message Queuing Protocol
b. Anchored Message Queuing Protocol
c. Advanced Message Queuing Protocol
d. None of these

Correct Answer: c. Advanced Message Queuing Protocol

Detailed Solution: AMQP stands for Advanced Message Queuing Protocol.

See lecture 8 (Basics of IoT Networking – Part IV) @ 0:41

QUESTION 11:

What is the basic unit of data in the AMQP protocol?

a. Chunk
b. Byte
c. Frame
d. None of these

Correct Answer: c. Frame

Detailed Solution: In AMQP, the basic unit of data is called a frame.

See lecture 8 (Basics of IoT Networking – Part IV) @ 0:41

QUESTION 12:

Which of the following is/are exchange types in AMQP?

a. Direct
b. Fan-out
c. Topic
d. All of these

Correct Answer: d. All of these

Detailed Solution: The AMQP exchange types are –

a) Direct
b) Fan-out
c) Topic
d) Header

See lecture 8 (Basics of IoT Networking – Part IV) @ 08:56

QUESTION 13:

IoT networks can be classified in which of the following type/types?

a. Non-Beacon Enabled
b. Beacon Enabled
c. None of these
d. Both (a) and (b)

Correct Answer: d. Both (a) and (b)

Detailed Solution: The IoT networks can be classified into two types such as

i. Non-Beacon Enabled
ii. Beacon Enabled

See lecture 9 (Connectivity Technologies – Part-I) @ 10:19

QUESTION 14:

The OSI model has _____ layers.

a. 6
b. 7
c. 4
d. 8

Correct Answer: b. 7

Detailed Solution: The OSI model is a conceptual framework that divides any networked communication system into seven layers.

See Page number – 10, Chapter - 1, Book - Introduction to IoT, Authors – Sudip Misra, Anandarup Mukherjee, and Arijit Roy, Publisher – Cambridge University Press, Edition – 1 (2021)

QUESTION 15:

The “Destination Address” in the IPv4 packet represents which of the following?

a. The destination node address of the packet
b. The intermediate hop in the network
c. Both (a) and (b)
d. None of these

Correct Answer: a. The destination node address of the packet

Detailed Solution: The “Destination Address” in the IPv4 packet represents the address of the destination node in the network.

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Introduction to Internet of Things

Assignment-Week 3

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

State whether the following statement is true or false.

Statement: There is no network layer in Wired HART

a. True
b. False

Correct Answer: a. True

Detailed Solution: There is no network layer in Wired HART.

See lecture 11 (Connectivity Technologies-III) @ 05:22

QUESTION 2:

HART operates only in _____ GHz ISM band.

a. 2.4
b. 4.8
c. 4.9
d. 4.4

Correct Answer: a. 2.4

Detailed Solution: HART operates only in 2.4 GHz ISM band.

See lecture 11 (Connectivity Technologies-III) @ 05:43

QUESTION 3:

Super-frames in HART consist of grouped _____ wide timeslots.

a. 10ms
b. 20ms
c. 55ms
d. None of these

Correct Answer: a. 10ms

Detailed Solution: Super-frames in HART consist of grouped 10ms wide timeslots.

See lecture 11 (Connectivity Technologies-III) @ 07:05

QUESTION 4:

The process of identifying channels consistently affected by interference and removing them from use is known as _____.

a. Channel hopping
b. Channel aggregating
c. Frequency aggregating
d. Channel blacklisting

Correct Answer: d. Channel blacklisting

Detailed Solution: Channel Blacklisting identifies channels consistently affected by interference and removes them from use.

See lecture 11 (Connectivity Technologies-III) @ 07:30

QUESTION 5:

State whether the following statement is True or False

Statement: Each node is supervised by the application manager and it guides them on when and where to send packets.

a. True
b. False

Correct Answer: b. False

Detailed Solution: The Network manager supervises each node in the network and guides them on when and where to send packets.

See lecture 11 (Connectivity Technologies-III) @ 12:14

QUESTION 6:

FeliCa is commonly found in which country?

a. Japan
b. India
c. USA
d. None of these

Correct Answer: a. Japan

Detailed Solution: FeliCa is commonly found in Japan.

See lecture 11 (Connectivity Technologies-III) @ 17:52

QUESTION 7:

State whether the following state is true or false.

Statement: NFC tags found in supermarket products are examples of passive NFC.

a. True

b. False

Correct Answer: a. True

Detailed Solution: Passive NFC tags found in supermarket products are examples of passive NFC.

See lecture 11 (Connectivity Technologies-III) @ 18:47

QUESTION 8:

Active NFC devices are able to _____?

a. transmit information
b. collect information
c. Both (a) and (b)
d. None of these

Correct Answer: c. Both (a) and (b)

Detailed Solution: Active NFC devices are able to collect as well as transmit information.

See lecture 11 (Connectivity Technologies-III) @ 18:52

QUESTION 9:

A _____ in NFC emits a small electric current which creates a magnetic field that in turn bridges the physical space between the devices.

a. reader
b. writer
c. destroyer
d. None of these

Correct Answer: a. reader

Detailed Solution: A reader in NFC emits a small electric current which creates a magnetic field that in turn bridges the physical space between the devices

See lecture 11 (Connectivity Technologies-III) @ 20:31

QUESTION 10:

Bluetooth technology is based on _____.

a. WiFi
b. LoRa
c. Ad-hoc technology
d. None of these

Correct Answer: c. Ad-hoc technology

Detailed Solution: Bluetooth technology is based on Ad-hoc technology.

See lecture 12 (Connectivity Technologies-IV) @ 02:30

QUESTION 11:

Bluetooth technology operates in the unlicensed industrial, scientific and medical (ISM) band at _____ to _____ GHz.

a. 3.2, 5.6
b. 2.4, 2.485
c. 5.0, 5.89
d. None of these

Correct Answer: b. 2.4,2.485

Detailed Solution: Bluetooth technology operates in the unlicensed industrial, scientific and medical (ISM) band at 2.4 to 2.485 GHz.

See lecture 12 (Connectivity Technologies-IV) @ 04:39

QUESTION 12:

Z-wave operates at _____ MHz in the US and _____ MHz in Europe.

a. 1008, 989
b. 908.42, 868.42
c. 767.56, 855.28
d. None of these

Correct Answer: b. 908.42, 868.42

Detailed Solution: Z-wave operates at 908.42 MHz in the US and 868.42 MHz in Europe.

See lecture 13 (Connectivity Technologies-V) @ 3:40

QUESTION 13:

State whether the following statement is true or false.

Statement: Sensor nodes have limited battery life.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Sensors nodes have limited battery life.

See lecture 14 (Sensor Networks-I) @ 13:15

QUESTION 14:

State whether the following statement is true or false.

Statement: The Link Manager Protocol in Bluetooth manages the establishment, authentication, link configuration.

a. True
b. False

Correct Answer: a. True

Detailed Solution: The Link Manager Protocol in Bluetooth manages the establishment, authentication, link configuration.

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QUESTION 15:

Zigbee commonly uses _____data rate.

a. 250 kbps
b. 250 Mbps
c. 260 kbps
d. 260 Mbps

Correct Answer: a. 250 kbps

Detailed Solution: Zigbee commonly uses 250 kbps data rate.

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**********END**********

Introduction to Internet of Things

Assignment-Week 4

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

State whether the following statement is true or false.

Statement: Coverage in WSN is defined as the area-of-interest that is covered satisfactorily.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Coverage in WSN is defined as the area-of-interest that is covered satisfactorily.

See lecture 17 (Sensor Networks-IV) @ 01:44

QUESTION 2:

If transmission range >= 2* sensing range,

a. Coverage implies connectivity
b. Coverage implies greater coverage
c. Both (a) and (b)
d. None of these

Correct Answer: a. Coverage implies connectivity

Detailed Solution: If transmission range >= 2* sensing range, coverage implies connectivity.

See lecture 17 (Sensor Networks-IV) @ 04:14

QUESTION 3:

Forest fire monitoring in WSN is an example of which type of reporting?

a. Event driven
b. On-demand
c. Scattered
d. None of these

Correct Answer: a. Event driven

Detailed Solution: Forest fire monitoring in WSN is an example of event driven reporting.

See lecture 17 (Sensor Networks-IV) @ 08:13

QUESTION 4:

State whether the following statement is true or false.

Statement: Objective of coverage in WSN is to use minimum number of sensors and maximize network lifetime.

a. True

b. False

Correct Answer: a. True

Detailed Solution: The objective of coverage in WSN is to use minimum number of sensors and maximize the network lifetime.

See lecture 17 (Sensor Networks-IV) @ 08:47

QUESTION 5:

What is the full form of AUV in the context of mobile wireless sensor networks?

a. Aerial and Underwater Vehicle
b. Ambient and Underprivileged Vehicle
c. Autonomous Underwater Vehicle
d. None of these

Correct Answer: c. Autonomous Underwater Vehicle

Detailed Solution: The full form of AUV is Autonomous Underwater Vehicle.

See lecture 18 (Sensor Networks-V) @ 07:56

QUESTION 6:

In mobile wireless sensor networks, the sensor nodes are capable of _____ and measuring the condition of their surrounding environments.

a. collaborating with one another
b. destroying one another
c. Both (a) and (b)
d. None of these

Correct Answer: a. collaborating with one another

Detailed Solution: In mobile wireless sensor networks, the sensor nodes are capable of collaborating with one another and measuring the condition of their surrounding environments.

See lecture 18 (Sensor Networks-V) @ 04:56

QUESTION 7:

State whether the following statement is true or false.

Statement: In mobile sensor networks, the sensor nodes come in close proximity of sink and deliver the data.

a. True
b. False

Correct Answer: a. True

Detailed Solution: In mobile sensor networks, the sensor nodes come in close proximity of sink and deliver the data.

See lecture 18 (Sensor Networks-V) @ 06:19

QUESTION 8:

Which of the following is/are a challenge/challenges in Human-centric Sensing?

a. Energy of devices
b. Participant selection
c. Both (a) and (b)
d. None of these

Correct Answer: c. Both (a) and (b)

Detailed Solution: The following are the challenges in human-centric sensing -

a. Energy of devices
b. Participant selection
c. Privacy of users

See lecture 18 (Sensor Networks-V) @ 12:45

QUESTION 9:

Which of the following is/are a feature/features of UAV networks?

a. Mesh or star networks
b. Multi-tasking
c. Large coverage area
d. All of these

Correct Answer: d. All of these

Detailed Solution: The following are the features of UAV networks -

d. Mesh or star networks
e. Multi-tasking
f. Large coverage area

See lecture 19 (UAV Networks) @ 02:43

QUESTION 10:

State whether the following statement is true or false.
Statement: Frequently change in network topology is a benefit of UAV networks.

a. False
b. True

Correct Answer: a. False

Detailed Solution: Frequently change in network topology an issue of UAV networks.

See lecture 19 (UAV Networks) @ 05:29

QUESTION 11:

Which of the following is NOT a feature of multi-UAV system?

a. Low failures

b. High scalability
c. Low survivability
d. High cost

Correct Answer: c. Low survivability

Detailed Solution: Following are some of the features of multi-UAV system –

a. Low failures
b. High scalability
c. High survivability
d. High cost

See lecture 19 (UAV Networks) @ 07:53

QUESTION 12:

State whether the following statement is true or false.

Statement: M2M communications are free of any human intervention.

a. True
b. False

Correct Answer: a. True

Detailed Solution: M2M communications are free of any human intervention.

See lecture 20 (Machine to Machine Communication) @ 05:44

QUESTION 13:

Which of the following is NOT an M2M node type?

a. Secure End Node
b. High End Node
c. Low End Mode
d. Mid End Node

Correct Answer: a. Secure End Node

Detailed Solution: M2M node types are as follows –

a. Low End Node

b. Mid End Node

c. High End Node

See lecture 20 (Machine to Machine Communication) @ 10:43

QUESTION 14:

What is the full form of M2SP in the context of Machine-to-Machine Communication?

a. M2M Service Platform
b. M2M Session Packet
c. M2M Secure Packaging
d. None of these

Correct Answer: a. M2M Service Platform

Detailed Solution: M2SP stands for M2M Service Platform.

See lecture 20 (Machine to Machine Communication) @ 16:31

QUESTION 15:

The _____ provides integrated services based on device collected data-sets.

a. M2M Application Platform
b. M2M Service Platform
c. M2M Hardware Platform
d. None of these

Correct Answer: a. M2M Application Platform

Detailed Solution: The M2M Application Platform provides integrated services based on device collected data-sets.

See lecture 20 (Machine to Machine Communication) @ 18:00

**************END*************

Introduction to Internet of Things

Assignment-Week 5

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Which of the following is/are current challenge/challenges in IoT?

a. Large scale of co-operation

b. Global heterogeneity

c. Unknown IoT device configuration

d. All of these

Correct Answer: d. All of these

Detailed Solution: Current challenges in IoT are –

a. Large scale of co-operation

b. Global heterogeneity

c. Unknown IoT device configuration

d. Semantic conflicts

See lecture 21 (Interoperability in Internet of Things) @ 03:41.

QUESTION 2:

When two IoT devices wish to communicate, semantic conflict in such a scenario in IoT interoperability refers to_____.

a. Two devices built by the same manufacturer

b. Two devices having different processing and business logic

c. Two devices sensing the same physical parameter

d. Two devices having different deployment location

Correct Answer: b. Two devices having different processing and business logic

Detailed Solution: Semantic conflict refers to when different IoT devices have different processing and business execution logic. Refer Lecture 21@5:32

QUESTION 3:

Which of the following issues needs to be addressed while solving user interoperability?

a. Device characterization and identification

b. Syntactic interoperability

c. Both (a) and (b)

d. None of these

Correct Answer: c. Both (a) and (b)

Detailed Solution: All of the options given should be addressed while solving user interoperability issues. Refer Lecture 21@15:16.

QUESTION 4:

State whether the following statement is true or false

Statement: UMB core interoperability component is responsible for converting physical devices into virtually abstracted ones.

a. False

b. True

Correct Answer: a. False

Detailed Solution: UMB-A is responsible for converting physical devices into virtually abstracted ones. (Please refer Lecture 21@27:16)

QUESTION 5:

What is the full form of UMB in IoT interoperability?

a. Universal Middleware Bridge

b. Universal Main Bridge

c. Universal Main Bracket

d. None of these

Correct Answer: a. Universal Middleware Bridge

Detailed Solution: UMB stands for Universal Middleware Bridge.

Refer Lecture 21@22:16.

QUESTION 6:

Which of the following is NOT a function available in the Servo library for Arduino ?

a. kill()

b. destroy()

c. burn()

d. All of these

Correct Answer: d. All of these

Detailed Solution: The following are NOT functions of the Servo library for Arduino –

a. kill()

b. destroy()

c. burn()

The Refer Lecture 25@18:08.

QUESTION 7:

Arduino UNO Board is open source?

a. True

b. False

Correct Answer: a. True

Detailed Solution: An Arduino UNO board is an open source electronic programmable board.

Refer Lecture 22@05:21.

QUESTION 8:

Arduino Uno board accepts which type of input/inputs?

a. Analog
b. Digital
c. Both (a) and (b)
d. None of these

Correct Answer: c. Both (a) and (b)

Detailed Solution: Arduino UNO board accepts analog and digital signals as input.

Refer Lecture 22@05:21

QUESTION 9:

Arduino boards are based on which of the following microcontroller/microcontrollers.

a. ATMEGA328
b. ATMEGA32u4
c. Both (a) and (b)
d. None of these

Correct Answer: c. Both (a) and (b)

Detailed Solution: Arduino board are based on the following microcontrollers –

a. ATMEGA328
b. ATMEGA32u4
c. ATMEGA2560
d. AT91SAM3X8E

Refer Lecture 22@06:20

QUESTION 10:

What is the clock speed of the Arduino Uno board?

a. 16MHz
b. 17MHz
c. 8MHz
d. None of these

Correct Answer: a. 16MHz

Detailed Solution: The clock speed of Arduino Uno board is 16MHz.

Refer Lecture 22 @ 07:13.

QUESTION 11:

Consider the following Arduino sketch.

```
int r = 4;
int g = 5;
void setup(){

Serial.begin(9600);
pinMode(r,INPUT);
pinMode(g,__??__);
}

void loop()
{
  int val = analogRead(r);
  if( val > 10)
  {
      digitalWrite(g,HIGH);
  }
  else
  {
      digitalWrite(g,HIGH);
  }
  delay(500);
}
```

What must be inserted in the place of '??' within the second pinMode() function in void setup()?

a. INPUT

b. OUTPUT

c. None of these

d. Anyone of these is okay

Correct Answer: b. OUTPUT

Detailed Solution: Since pin 'g' is writing digital values, it is required to set in OUTPUT mode. Refer Lecture 22-23.

QUESTION 12:

The "Verify" option in the Arduino IDE checks the code for _____.

a. Compilation errors
b. Improper hardware connection
c. Both (a) and (b)
d. None of these

Correct Answer: a. Compilation errors

Detailed Solution: The "Verify" option in the Arduino IDE checks the code for compilation errors. Refer Lecture 24.

QUESTION 13:

How many pins does the DHT Digital Humidity and Temperature sensor have?

a. 2
b. 3
c. 4
d. None of these

Correct Answer: c. 4

Detailed Solution: DHT Digital Humidity and Temperature sensor has 4 pins.

See lecture 24 @ 03:23

QUESTION 14:

What is the function of the delay() function in an Arduino program?

a. Initializes a sensor
b. Introduces a delay for the specified time
c. Stores the sensor value
d. None of the above

Correct Answer: b. Introduces a delay for the specified time

Detailed Solution: The delay() function introduces a delay in the operation of the microcontroller for the specified duration.

See lecture 24 @13:51

QUESTION 15:

What does the following code snippet do?

Servo myservo;

a. Creates a variable to store the sensor value
b. Instructs the sensor to sense data
c. Creates an instance of servo to use it in the Arduino sketch
d. None of these

Correct Answer: c. Creates an instance of servo to use it in the Arduino sketch

Detailed Solution: The code snipped “Servo myservo;” creates an instance of servo to use in the Arduino sketch.

Refer Lecture 25 @5:32

**************END*************

Introduction to Internet of Things

Assignment-Week 6

TYPE OF QUESTION:MCQ/MSQ

Number ofquestions:15 **Total marks: 15 X 1= 15**

QUESTION 1:

Python can only be used to build small applications.

a. True

b. False

Correct Answer: b. False

Detailed Solution: With open-source nature, python forms a strong backbone to build large applications.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 3:46)

QUESTION 2:

Which of the following is NOT a Python IDE?

a. Spyder

b. PyCharm

c. Both (a) and (b)

d. None of these

Correct Answer: d. None of these

Detailed Solution: Spyder and PyCharm are examples of Python IDE.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 05:12)

QUESTION 3:

State whether the following statement is true or false.

Statement: To indicate different blocks of code, python follows rigid indention.

a. False

b. True

Correct Answer: b. True

Detailed Solution: To indicate different blocks of code, python follows rigid indention (Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 07:25).

QUESTION 4:

Which of the following is not a data-type in Python?

a. List

b. Branch

c. Tuple

d. None of these

Correct Answer: b. Branch

Detailed Solution: The data-types in python are –

a. List

b. Tuple

c. Dictionary

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- I @ 15:20).

QUESTION 5:

State whether the following statement is true or false.

Statement: Python allows us to read and write files.

a. True

b. False

Correct Answer: a. True

Detailed Solution: Python allows us to read and write files.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 01:17)

QUESTION 6:

Which of the following is NOT a mode to open a file?

a. Read mode

b. Write mode

c. Append mode

d. None of these

Correct Answer: d. None of these

Detailed Solution: The four basic modes to open a file in python are –

a. Read mode
b. Write mode
c. Append mode
d. Both read and write mode

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 03:00)

QUESTION 7:

State whether the following command to install the PIL library in Linux is correct or not.

sudo pip install pillow

a. Correct
b. Incorrect

Correct Answer: a. Correct

Detailed Solution: The command to install the PIL library is *sudo pip install pillow*.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 17:40)

QUESTION 8:

State whether the following statement is correct for creating a socket 's' in python.

s = socket.socket(socket_family, socket_type, protocol=0)

a. Correct
b. Incorrect

Correct Answer: a. Correct

Detailed Solution: The statement *s = socket.socket(socket_family, socket_type,*

protocol=0) creates a socket in python.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @24:27).

QUESTION 9:

What is a Raspberry Pi?

a. A computer on your palm
b. Single board computer
c. Low cost
d. All of these

Correct Answer: d. All of these

Detailed Solution: Raspberry pi is –

a. A computer on your palm
b. Single board computer
c. Low cost
d. Easy to access

(Please refer to lecture INTRODUCTION TO RASPBERRY PI-I @05:00).

QUESTION 10:

Which of the following are required for a basic set up of Raspberry Pi?

a. Monitor
b. Keyboard
c. Mouse
d. All of these

Correct Answer: d. All of these

Detailed Solution: The following are required for a basic set up of Raspberry Pi –

a. Monitor
b. Keyboard
c. Mouse

See lecture INTRODUCTION TO RASPBERRY PI-I @ 11:33

QUESTION 11:

Which of the following keyboard key combination writes (i.e saves) the code to the file in nano editor ?

a. Ctrl+O
b. Ctrl+X
c. Both (a) and (b)
d. None of these

Correct Answer: a. Ctrl+O

Detailed Solution: Ctrl+O writes the code to the file in nano editor.

See lecture INTRODUCTION TO RASPBERRY PI-II @ 10:44

QUESTION 12:

A relay module for Raspberry Pi has how many terminals ?

a. 3
b. 2
c. 4
d. 5

Correct Answer: a. 3

Detailed Solution: A relay module for Raspberry Pi has 3 terminals.

See lecture IMPLEMENTATION OF IOT WITH RASPBERRY PI-I @ 06:01.

QUESTION 13:

State whether the following statement is true or false.

Statement: Adafruit provides a library to work with the DHT22 sensor.

a. True

b. False

Correct Answer: a. True

Detailed Solution: Adafruit provides a library to work with the DHT22 sensor.

See lecture IMPLEMENTATION OF IOT WITH RASPBERRY PI-I @ 09:19

QUESTION 14:

What is the function of the close() function in python programming while handling files.

a. This function does not exist in python

b. It closes the current connection to the file and ensures that the file is free to use for other resources

c. None of these

d. Both (a) and (b)

Correct Answer: b. It closes the current connection to the file and ensures that the file is free to use for other resources

Detailed Solution: The close() function in python ensures that the file is free to use for other resources.

(Please refer to lecture INTRODUCTION TO PYTHON PROGRAMMING- II @ 4:44)

QUESTION 15:

Where is the Raspbian image found?

a. It is not available on the Internet

b. It is available as image file from the Raspberry Pi official website

c. It can only be purchased from offline vendor

d. None of these

Correct Answer: b. It is available as image file from the Raspberry Pi official website

Detailed Solution: Raspbian image can be downloaded from the Raspberry Pi official website.

(Please refer to lecture INTRODUCTION TO RASPBERYY PI-I @ 12:32)

**************END*************

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 7

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

DHT Sensor refers to Digital Humidity and Temperature Sensor.

a. True
b. False

Correct Answer: a. True
Detailed Solution: DHT Sensor refers to Digital Humidity and Temperature Sensor. (Please refer to lecture Implementation of IoT with Raspberry Pi- II @ 3:28)

QUESTION 2:

Fill in the blanks. _____ is a python library used for plotting the data in 2D.

a. List
b. Numpy
c. Pandas
d. Matplotlib

Correct Answer: d. Matplotlib

Detailed Solution: Matplotlib is a python library used for plotting the data in 2D. (Please refer to lecture Implementation of IoT with Raspberry Pi- III @ 12:13)

QUESTION 3:

Fill in the blanks. _____ makes a scatter plot of the given points.

a. ion()
b. figure()
c. Scatter()
d. None of these

Correct Answer: c. Scatter()

Detailed Solution: Scatter() makes a scatter plot of the given points. (Please refer to lecture Implementation of IoT with Raspberry Pi- III @ 23:25).

QUESTION 4:

What are the two main challenges of SDN?

a. File placement and Node placement
b. Rule placement and Controller placement
c. All of these
d. None of these

Correct Answer: b. Rule placement and Controller placement

Detailed Solution: Rule placement and Controller placement are the two main challenges of SDN. (Please refer to lecture Software Defined Networking- Part- I @ 16:49)

QUESTION 5:

In soft-timeout, all the rules are deleted from the switch.

a. True
b. False

Correct Answer: b. False

Detailed Solution: In soft-timeout, if no flow is received associated with a rule for a particular time, the rule is deleted. In hard-timeout, all rules are deleted from the switch. (Please refer to lecture Software Defined Networking- Part- I @ 24:57)

QUESTION 6:

Which of the following is a component of SDN?

a. Hardware switches
b. Flow-rules
c. None of these
d. All of these

Correct Answer: d. All of these

Detailed Solution: Hardware switches and Flow rules are both the components of SDN. (Please refer to lecture Software Def ined Networking- Part- I @ 15.47)

QUESTION 7:

What is introduced if a controller is down in SDN?

a. Backup controller
b. Intro controller
c. All of these
d. None of these

Correct Answer: a. Backup Controller

Detailed Solution: Backup controller is introduced if a controller is down in SDN. (Please refer to lecture Software Defined Networking- Part- II @ 8:29)

QUESTION 8:

Southbound API is used to communicate between control layer and application layer.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Southbound API is used to communicate between control layer and infrastructure layer. (Please refer to lecture 34 Software Defined Networking- Part- II @1:41).

QUESTION 9:

SDN is OpenFlow.

a. True
b. False

Correct Answer: b. False

Detailed Solution: SDN is not OpenFlow. It is a technology/concept. (Please refer to lecture 33 Software Defined Networking- Part- I @25:55).

QUESTION 10:

Which of the following forwards the sensed data based on the ID of the source node?

a. Value-centric data forwarding
b. ID-centric data forwarding
c. All of these
d. None of these

Correct Answer: b. ID-centric data forwarding

Detailed Solution: ID-centric data forwarding forwards the sensed data based on the ID of the source node. See lecture 35 @ 9:50

QUESTION 11:

Does integrating SDN in IoT provide intelligent routing decisions?

a. Yes
b. No

Correct Answer: a. Yes

Detailed Solution: Integrating SDN in IoT provides intelligent routing decisions. See lecture 35 @ 17:25

QUESTION 12:

Are Indigo and LINC Open source?

a. Yes
b. No

Correct Answer: a. Yes

Detailed Solution: Indigo and LINC are both open source. See lecture 33@ 26:43.

QUESTION 13:

Which of the following is used to communicate among multiple controllers in the control layer?

a. East-Westbound APIs
b. Northbound APIs
c. All of these
d. None of these

Correct Answer: a. East-Westbound APIs

Detailed Solution: East-Westbound APIs are used to communicate among multiple controllers in the control layer.
See lecture 34 @ 02:39

QUESTION 14:

How many requests can a controller handle through a single thread?

a. 200/sec
b. 50/sec
c. None of these
d. All of these

Correct Answer: a. 200/sec

Detailed Solution: A controller can handle 200 requests in a second through a single thread. (Please refer to lecture 34@ 3:47)

QUESTION 15:

Size of the ternary content-addressable memory is not limited at the switches.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Size of the ternary content-addressable memory is limited at the switches (Please refer to lecture 33 @ 20:41)

**************END*************

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 8

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Which of these is used for efficient base-station coordination for addressing inter-cell interference?

a. Path management
b. Routing protocols
c. Flow-table paradigm
d. Logically centralized control

Correct Answer: d. Logically centralized control

Detailed Solution: Logically centralized control is used for efficient base-station coordination for addressing inter-cell interference (Please refer Lecture 36@3:02)

QUESTION 2:

Which of these takes last k-th location instances to predict the next location in Mobi-Flow?

a. Order-K Markov predictor

b. Flow-rule placement

c. All of these

d. None of these

Correct Answer: a. Order-K Markov predictor

Detailed Solution: Order-K Markov predictor takes last k-th location instances to predict the next location in Mobi-Flow. (Please refer Lecture 36@15:52)

QUESTION 3:

What are the two components of ODIN?

a. Odin server and agent
b. Odin server and client
c. Odin agent and master
d. None of these

Correct Answer: c. Odin agent and master

Detailed Solution: Odin agent and master are the two components of ODIN. (Please refer Lecture 36@10:16)

QUESTION 4:

Which of the following is placed to deal with mice-flows ?

a. Flow rules
b. Exact rules
c. Match rules
d. Wild card rules

Correct Answer: d. Wild card rules

Detailed Solution: Wild card rules can be placed to deal with mice-flows. (Please refer Lecture 36 @18:23)

QUESTION 5:

Fill in the blank.

_____ refers to heterogeneous computing nodes distributed over a wide area to perform very large tasks.

a. Cluster Computing
b. Utility Computing
c. Grid Computing
d. None of these

Correct Answer: c. Grid Computing

Detailed Solution: Grid Computing refers to heterogeneous computing nodes distributed over a wide area to perform very large tasks. (Please refer Lecture 37 @6:48)

QUESTION 6:

Which of the following is a property of cloud computing?

a. On-demand network access
b. High level generalization of computation
c. On-demand services
d. All of these

Correct Answer: d. All of these

Detailed Solution: On-demand network access, high level generalization of computation and on-demand services are some of the properties of cloud computing. (Please refer Lecture 37@12:43)

QUESTION 7:

Which of the following statements are true about the different types of cloud?

Statement I: Inter-cloud is a global cloud of clouds.

Statement II: Volunteer cloud is a type of distributed cloud.

Statement III: Public cloud has low security.

a. Statements I and II
b. Statements I and III
c. Statements II and III
d. Statement I, II and III

Correct Answer: d. Statements I, II and III

Detailed Solution: All the three statements are correct for the cloud. Inter-cloud is a global cloud of clouds. Volunteer cloud is a type of distributed cloud. And also Public cloud has low security. (Please refer Lecture 37@31:11, 32: 43, 32: 25)

QUESTION 8:

Fill in the blank.

_____ means independent of device or location.

a. Scalable
b. Reliability
c. Agile
d. Ubiquitous

Correct Answer: d. Ubiquitous

Detailed Solution: Ubiquitous means independent of device or location. (Please refer Lecture 37@18:06)

QUESTION 9:

Fill in the blank.

_____ represents the on-demand delivery of computing infrastructure in the cloud.

a. Software-as-a-Service
b. Storage-as-a-Service
c. Network-as-a-Service
d. Infrastructure-as-a-Service

Correct Answer: d. Infrastructure-as-a-Service

Detailed Solution: Infrastructure-as-a-Service represents the on-demand delivery of computing infrastructure in the cloud. (Please refer Lecture 38@4:29)

QUESTION 10:

Fill in the blank. Hybrid cloud is a blend of _____ cloud.

a. Public and private

b. SaaS and Paas

c. All of these

d. None of these

Correct Answer: a. Public and private

Detailed Solution: Hybrid cloud is a blend of public and private cloud. (Please refer Lecture 38@9:50)

QUESTION 11:

Software-as-a-Service does not support a fully pay-as-you-go model.

a. True
b. False

Correct Answer: b. False

Detailed Solution: Software-as-a-Service supports a fully pay-as-you-go model. (Please refer Lecture 38@13:25)

QUESTION 12:

Which of the following describes the quality of services expected by the customer?

a. Mass production
b. Internet evolution
c. Automation
d. Service Level Agreement

Correct Answer: d. Service Level Agreement

Detailed Solution: Service Level Agreement describes the quality of services expected by the customer (Please refer Lecture 39@5:23)

QUESTION 13:

Which of the following are the accesses enabled by the access control layers in the cloud?

a. Service access
b. Server access
c. Database access
d. All of these

Correct Answer: d. All of these

Detailed Solution: Service access, Server access and Database access are the accesses enabled by the access control layers in the cloud (Please refer Lecture 39@24:31)

QUESTION 14:

Reputation refers to the belief of an entity's standing by the community.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Reputation refers to the belief of an entity's standing by the community. (Please refer Lecture 39@25:15)

QUESTION 15:

Which of the following is true about GreenCloud?

a. GreenCloud monitors the energy consumption of servers, switches, etc.
b. GreenCloud was developed as an extension of NS2.
c. All of these
d. None of these

Correct Answer: c. All of these

Detailed Solution: GreenCloud monitors the energy consumption of servers, switches, etc. It was developed as an extension of NS2. (Please refer Lecture 40@16:25)

**********END**********

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 9

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

Which of the following is not a component of OpenStack?

a. Suse
b. Nova
c. Swift
d. All of these

Correct Answer: a. Suse

Detailed Solution: Nova and Swift are two of the many components of OpenStack (Please refer Lecture 41@3:18)

QUESTION 2:

Where is the sensed data sent for further processing in WSNs?

a. Client node
b. Central node
c. Collect node
d. Sink node

Correct Answer: d. Sink node

Detailed Solution: The sensed data is sent to sink node for further processing in WSNs (Please refer Lecture 42@4:04)

QUESTION 3:

Select the statement(s) that denote the advantages of cloud computing.

Statement I: Elasticity

Statement II: Pay-per-use

Statement III: Self Service

a. Statement I

b. Statement II

c. Statements I, II, and III

d. None of these

Correct Answer: c. Statements I, II, and III

Detailed Solution: The advantages of cloud computing are Elasticity, pay-per-use and self service (Please refer Lecture 42@5:58)

QUESTION 4:

Which of the following is a limitation of WSN?

a. Price

b. Post deployment maintenance

c. Battery lifetime

d. All of these

Correct Answer: d. All of these

Detailed Solution: Price, post deployment maintenance and battery lifetime are the main

limitations of WSN. (Please refer Lecture 42@11:01)

QUESTION 5:

The managerial role is played by _____ in sensor-cloud architecture.

a. End-users

b. Sensor-Cloud Service Provider

c. None of these

d. All of these

Correct Answer: b. Sensor-Cloud Service Provider

Detailed Solution: Sensor-Cloud Service Provider plays the managerial role in sensor-cloud architecture. (Please refer Lecture 42@14:29)

QUESTION 6:

The optimal composition of _____ is a management issue in sensor-cloud.

a. Logistics

b. Pricing

c. Caching

d. Virtual sensor nodes

Correct Answer: d. Virtual sensor nodes

Detailed Solution: The optimal composition of Virtual sensor nodes is a management issue in sensor-cloud. (Please refer Lecture 43@4:39)

QUESTION 7:

Who coined the term Fog computing?

a. IBM

b. CISCO

c. All of these

d. None of these

Correct Answer: b. CISCO

Detailed Solution: CISCO coined the term Fog computing. (Please refer Lecture 44@4:18)

QUESTION 8:

What are the issues related to the current cloud model?

a. Volume

b. Latency

c. All of these

d. None of these

Correct Answer: c. All of these

Detailed Solution: Volume, bandwidth and latency are some of the issues related to the current cloud model. (Please refer Lecture 44@10:30)

QUESTION 9:

Sensor data is processed in the cloud before it is sent to the fog.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Sensor data is processed in the fog before it is sent to the cloud. (Please refer Lecture 45@6:11)

QUESTION 10:

What is the type of received data according to which Fog nodes work?.

a. Non-time sensitive data
b. Very time-sensitive data
c. Less time-sensitive data
d. All of these

Correct Answer: d. All of these

Detailed Solution: Fog nodes work according to the type of data which are very time-sensitive, less time-sensitive and non-time sensitive data. (Please refer Lecture 45@11:39)

QUESTION 11:

Which of the following are the advantages of Fog?

a. Quick decision making

b. Nodes can be mobile

c. Reduces the risk of latency

d. All of these

Correct Answer: d. All of these

Detailed Solution: The advantages of Fog includes quick decision making, nodes

can be mobile and Fog reduces the risk of latency . (Please refer Lecture 45@20:57)

QUESTION 12:

Which of the following requires the analysis at the nearest node?

a. Non-time sensitive data

b. Most time-sensitive data

c. Less time-sensitive data

d. None of these

Correct Answer: b. Most time-sensitive data

Detailed Solution: Most time-sensitive data requires the analysis at the nearest node. (Please refer Lecture 45@15:46)

QUESTION 13:

Fog nodes cannot connect and leave the network when necessary.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Fog nodes can connect and leave the network when necessary. (Please refer Lecture 45@25:27)

QUESTION 14:

Which of the following is used by the fog applications to increase the speed of service accessibility?

a. Low power

b. Good network connection

c. Accelerators

d. Analytics

Correct Answer: c. Accelerators

Detailed Solution: Accelerators increase the speed of service accessibility. (Please refer to Page 260, Chapter 11, Introduction to IoT. S. Misra, A. Mukherjee, and A. Roy, 2020. Cambridge University Press.)

QUESTION 15:

Fill in the blank. The concept of _____ enables physical hardware to be shared among multiple entities.

a. Hardware virtualization

b. Software virtualization

c. Module virtualization

d. All of these

Correct Answer: a. Hardware virtualization

Detailed Solution: The concept of hardware virtualization enables physical hardware to be shared among multiple entities. (Please refer to Page 262, Chapter 11, Introduction to IoT. S. Misra, A. Mukherjee, and A. Roy, 2020. Cambridge University Press.)

**************END*************

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 10

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

Smart homes can be used for _____.

a. Home monitoring
b. Conservation of resources like electricity
c. Security and safety
d. All of these

Correct Answer: d. All of these

Detailed Solution: Smart homes can be used for home monitoring, conservation of resources, security and safety. (Please refer Lecture 46@25:02)

QUESTION 2:

Which of the following is an iot challenge in smart cities?

a. Reliability
b. Large scale
c. Big data
d. All of these

Correct Answer: d. All of these

Detailed Solution: Reliability, Large scale and Big data are some of the challenges of iot in smart cities. (Please refer Lecture 46@32:57)

QUESTION 3:

Select the statement(s) that are true for smart parking.

Statement I: Smart parking reduces traffic congestion

Statement II: Smart parking increases urban mobility

Statement III: Smart parking reduces fuel consumption

a. Statement I

b. Statement II

c. Statements I, II, and III

d. None of these

Correct Answer: c. Statements I, II, and III

Detailed Solution: Smart parking reduces traffic congestion and fuel consumption, and increases urban mobility (Please refer Lecture 47@16:46)

QUESTION 4:

Which of the following is a functional layer in smart parking?

a. Information Collection

b. System Deployment

c. Service Dissemination

d. All of these

Correct Answer: d. All of these

Detailed Solution: Information Collection, System Deployment and Service Dissemination are

all the functional layers in smart parking. (Please refer Lecture 47@17:24)

QUESTION 5:

The mobile sensing in smart parking collects the information _____.

a. in-place

b. along the route

c. None of these

d. All of these

Correct Answer: b. along the route

Detailed Solution: The mobile sensing in smart parking collects the information along the route. (Please refer Lecture 47@19:47)

QUESTION 6:

The network which provides amalgamation of various systems within a home is known as _____.

a. Local Area Network

b. Metropolitan Area Network

c. Global Area Network

d. Home Area Network

Correct Answer: d. Home Area Network

Detailed Solution: The network which provides amalgamation of various systems within a home is known as Home Area Network. (Please refer Lecture 48@8:24)

QUESTION 7:

Which of the following is the correct full form of DLNA Standards for HAN?

a. Digital Lifestyle Network Alliance

b. Digital Living Network Alliance

c. Digital Lifestyle Network Algorithm

d. None of these

Correct Answer: b. Digital Living Network Alliance

Detailed Solution: DLNA stands for Digital Living Network Alliance standards. (Please refer Lecture 48@13:49)

QUESTION 8:

Which of the following is a part of ZigBee Alliance?

a. Mitsubishi

b. Honeywell

c. All of these

d. None of these

Correct Answer: c. All of these

Detailed Solution: The Zigbee Alliance is composed of Mitsubishi, Honeywell, Invensys, Motorola and Philips. (Please refer Lecture 48@15:46)

QUESTION 9:

Network and Application layers are defined by IEEE 802.15.4 in Zigbee.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Network and Application layers are defined by Zigbee in Zigbee. (Please refer Lecture 48@16:42)

QUESTION 10:

Which of the following is a feature of Content Centric Networking?

a. Hierarchically named data
b. Scalable data dissemination
c. In-network caching
d. All of these

Correct Answer: d. All of these

Detailed Solution: Hierarchically named data, Scalable data dissemination and In-network caching are all the features of CCN. (Please refer Lecture 49@15:17)

QUESTION 11:

Which of the following is a phase of ICV development?

a. Based on 2G

b. Based on 4G LTE

c. Vehicles connected to cloud

d. All of these

Correct Answer: d. All of these

Detailed Solution: The phases of ICV development: Phase 1: Based on 2G, Phase 2: Based on 4G LTE, Phase 3: Vehicles connected to cloud (Please refer Lecture 50@12:28)

QUESTION 12:

Which of the following is a property of V2X?

a. Restricted to a single source

b. Distributed architecture

c. Both of these

d. None of these

Correct Answer: b. Distributed architecture

Detailed Solution: V2X follows a distributed architecture. (Please refer Lecture 49@12:06)

QUESTION 13:

The vehicles cannot be mobile in the Ad-hoc domain of VANET.

a. True

b. False

Correct Answer: b. False

Detailed Solution: The vehicles are mobile in the Ad-hoc domain of VANET.. (Please refer Lecture 50@16:59)

QUESTION 14:

Which of the following is a domain of VANET?

a. In-vehicle

b. Ad-hoc

c. All of these

d. None of these

Correct Answer: c. All of these

Detailed Solution: Both of the In-vehicle and Ad-hoc are the two domains of VANETs. (Please refer Lecture 50@14:36)

QUESTION 15:

Fill in the blank. ICV is based on _____.

a. Dedicated Short Range Communication

b. Directed Short Range Communication

c. Dedicated Small Range Communication

d. All of these

Correct Answer: a. Dedicated Short Range Communication

Detailed Solution: ICV is based on Dedicated Short Range Communication.

(Please refer Lecture 50@7:22)

**************END*************

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 11

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **15** **Total marks: 15 X 1= 15**

QUESTION 1:

What are some of the other names of the smart grids?

a. Electricity with a brain
b. Energybar
c. Electrolyte
d. All of these

Correct Answer: a. Electricity with a brain

Detailed Solution: Smart grids are also known as Electricity with a brain, Energy internet, and Electronet. (Please refer Lecture 51@8:33)

QUESTION 2:

Traditional Grid follows which type of flow of energy?

a. Unidirectional

b. Bidirectional

c. Both a and b

d. None of these

Correct Answer: a. Unidirectional

Detailed Solution: Traditional grid follows unidirectional flow of energy. (Please refer Lecture 51@5:41)

QUESTION 3:

Which of the following is a benefit associated with smart grids?

a. Efficient transmission of electricity
b. Lower electricity rates
c. Improved security
d. All of these

Correct Answer: d. All of these

Detailed Solution: Smart Grids provide efficient transmission of electricity, lower electricity rates and Improved security. (Please refer Lecture 51@11:26)

QUESTION 4:

Which of the following is a way of consumer engagement?

a. Net metering
b. Time-of-Use pricing
c. Financial incentives
d. All of these

Correct Answer: d. All of these

Detailed Solution: Net metering, Time-of-Use pricing and Financial incentives are ways of consumer engagements. (Please refer Lecture 51 @32:31)

QUESTION 5:

Fill in the blank.

In _____, a home can have power from distributed resources.

a. Peaking
b. Islanding
c. Neither a nor b

d. Both a and b

Correct Answer: b. Islanding

Detailed Solution: In islanding, a home can have power from distributed resources.(Please refer Lecture 51 @31:20)

QUESTION 6:

Which of the following aggregates energy consumption of a certain geographical area?

a. Data Aggregator Units
b. Meter Data Management System
c. Buffer

d. None of these

Correct Answer: a. Data Aggregator Units

Detailed Solution: Data Aggregator Units aggregates energy consumption of a certain geographical area (Please refer Lecture 52@15:03)

QUESTION 7:

Which of the following statements are true about Dynamic System Attacks?

Statement I: Replay attacks compromises sensors and monitors their outputs.

Statement II: D-DIA can lead to a system collapse.

Statement III: Covert attack is a closed loop version of replay attacks.

a. Statements I and II
b. Statements I and III
c. Statements II and III
d. Statement I, II and III

Correct Answer: d. Statements I, II and III

Detailed Solution: Replay attacks compromises sensors and monitors their outputs. D-DIA can lead to a system collapse. Covert attack is a closed loop version of replay attacks. (Please refer Lecture 52@20:03)

QUESTION 8:

Fill in the blank.

_____ is a centralized coordinator for smart grid communication.

a. Gateway
b. Meter
c. AMH
d. MDMS

Correct Answer: d. MDMSs

Detailed Solution: MDMS is a centralized coordinator for smart grid communication. (Please refer Lecture 52@15:32)

QUESTION 9:

Fill in the blank.

_____ are available protocols for smart home appliances.

a. C-Bus
b. DECT
c. EnOcean
d. All of these

Correct Answer: d. All of these

Detailed Solution: C-Bus, DECT, EnOcean and Universal Powerline Bus are available protocols for smart home appliances. (Please refer Lecture 52@12:35)

QUESTION 10:

Fill in the blank. _____ is the time required to transmit the data.

a. Scalability

b. Topology

c. Throughput

d. Latency

Correct Answer: d. Latency

Detailed Solution: Latency is the time required to transmit the data. (Please refer Lecture 53@21:20)

QUESTION 11:

IIoT is based on Wrap & Replace approach.

a. True
b. False

Correct Answer: b. False

Detailed Solution: IIoT is based on wrap & re-use approach. (Please refer Lecture 53@9:53)

QUESTION 12:

Which of the following happened in the 3rd Revolution?

a. Mass production
b. IIoT
c. Mechanized production
d. None of these

Correct Answer: d. None of these

Detailed Solution: Internet evolution and automation happened in the 3rd Revolution.(Please refer Lecture 53@10:06)

QUESTION 13:

Which of the following is used to improve the interoperability of different systems?

a. Standardization
b. Optimization
c. Monitoring
d. None of these

Correct Answer: a. Standardization

Detailed Solution: Standardization improves the interoperability of different systems. (Please refer Lecture 54@16:09)

QUESTION 14:

Big data is characterized by 7 Vs.

a. True
b. False

Correct Answer: a. True

Detailed Solution: Big data is characterized by 7 Vs. (Please refer Lecture 55@10:36)

QUESTION 15:

What is Hadoop used in?

a. Distributed processing of large datasets
b. Small clusters of computers
c. Both a and b
d. None of these

Correct Answer: a. Distributed processing of large datasets

Detailed Solution: Hadoop is used for distributed processing of large datasets across Large clusters of computers (Please refer Lecture 55@23:18)

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**************END*************

NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Introduction to Internet of Things

Assignment-Week 12

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 15 **Total marks: 15 X 1= 15**

QUESTION 1:

The two types of data analysis are _____.

a. Qualitative and Quantitative
b. Repetitive and Quantitative
c. Neither a nor b
d. Both a and b

Correct Answer: a. Qualitative and Quantitative

Detailed Solution: Qualitative and Quantitative are the two types of data analysis. (Please refer Lecture 56@2:33)

QUESTION 2:

Which of the following is a principle of Qualitative analysis.

a. Notice things
b. Think about things
c. Collect things
d. All of these

Correct Answer: d. All of these

Detailed Solution: The three principles of Qualitative analysis are: Notice things, think about things and collect things. (Please refer Lecture 56@4:12)

QUESTION 3:

Select the statement(s) that is involved with quantitative analysis.

Statement I: Data dispersion

Statement II: Contingence

Statement III: Correlation

a. Statement I

b. Statement II

c. Statements I, II, and III

d. None of these

Correct Answer: c. Statements I, II, and III

Detailed Solution: Data dispersion, Contingence and Correlation are involved with quantitative analysis. (Please refer Lecture 56@4:46)

QUESTION 4:

Which of the following is a type of data dispersion?

a. Range
b. Average absolute deviation
c. Both a and b
d. Neither a nor b

Correct Answer: c. Both a and b

Detailed Solution: Range and average absolute deviation are the type of data dispersions. (Please refer Lecture 56@13:43)

QUESTION 5:

In ANOVA, if the distribution of each group is normal, it is known as _____.

a. Homogeneity
b. Normality
c. Neither a nor b
d. Both a and b

Correct Answer: b. Normality

Detailed Solution: In ANOVA, if the distribution of each group is normal, it is known as normality. (Please refer Lecture 56@11:01)

QUESTION 6:

The expectation of the squared deviation of a random variable from its mean is known as _____.

a. Absolute deviation

b. Range

c. Standard deviation

d. Variance

Correct Answer: d. Variance

Detailed Solution: The expectation of the squared deviation of a random variable from its mean is known as Variance. (Please refer Lecture 56@14:04)

QUESTION 7:

What are the two types of statistical models?

a. Regression and dispersion
b. Complete and incomplete
c. Both a and b
d. Neither a nor b

Correct Answer: b. Complete and incomplete

Detailed Solution: Complete and incomplete are two types of statistical models (Please refer Lecture 56@8:11)

QUESTION 8:

Which of the following is a type of table that displays the frequency distribution of the variables?

a. Cross tabulation
b. Contingency table
c. Both a and b
d. Neither a nor b

Correct Answer: c. Both a and b

Detailed Solution: Contingency table is a type of table that displays the frequency distribution of the variables. It is also known as Cross tabulation and Cross tab. (Please refer Lecture 56@14:29)

QUESTION 9:

Advances in sensor and connectivity have enabled preventive care in IoT Healthcare.

a. True

b. False

Correct Answer: a. True

Detailed Solution: Advances in sensor and connectivity have enabled preventive care in IoT Healthcare. (Please refer Lecture 58@4:22)

QUESTION 10:

Which of the following is an application of IoT in Agriculture?.

a. Automated irrigation system
b. Automated weeding system
c. Water level monitoring
d. All of these

Correct Answer: d. All of these

Detailed Solution: Automated irrigation system, Automated weeding system and Water level monitoring are all applications of IoT in agriculture. (Please refer Lecture 57@2:05)

QUESTION 11:

Which of the following is a property of in-place data analysis?

a. Power intensive

b. On the device

c. No need of network

d. All of these

Correct Answer: d. All of these

Detailed Solution: In-place data analysis is power intensive, can be done on the device and needs no network connection.

(Please refer Lecture 59@20:24)

QUESTION 12:

Which of the following is the advantage of activity monitoring?

a. Expensive
b. Long term monitoring
c. Both a and b
d. Neither a nor b

Correct Answer: b. Long term monitoring

Detailed Solution: Long term monitoring is one of the advantages of Activity monitoring. (Please refer Lecture 59@16:17)

QUESTION 13:

Deep learning based data analysis cannot be performed on sensor data.

a. True

b. False

Correct Answer: b. False

Detailed Solution: Deep learning based data analysis can be done on sensor data, images and videos. (Please refer Lecture 59@19:34)

QUESTION 14:

Which of the following is true for network-based data analysis?

a. Low power

b. Good network connection required

c. Group-based analytics

d. All of these

Correct Answer: d. All of these

Detailed Solution: Network-based data analysis uses low power, good network connection and can perform group-based analytics (Please refer Lecture 59@20:10)

QUESTION 15:

Which of the following is an advantage of continuous activity monitoring?

a. Daily observation of behavior
b. Long term monitoring
c. Neither a nor b
d. Both a and b

Correct Answer: d. Both a and b

Detailed Solution: Daily observation of behavior and Long term monitoring are two advantages of continuous activity monitoring. (Please refer Lecture 59@16:15)

**************END*************